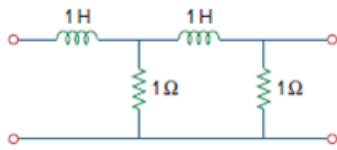


Problem

For the ladder network in Fig. 19.30, determine the g parameters in the s domain.

Figure 19.30



Step-by-step solution

Step 1 of 9

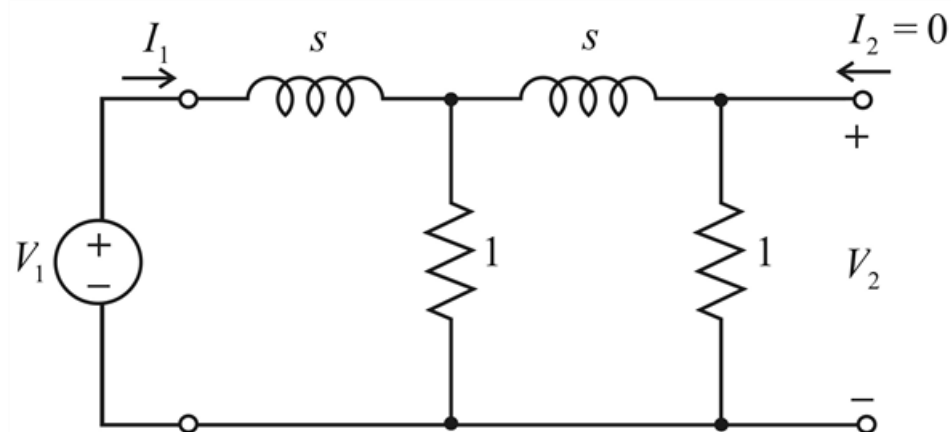
The g parameters are

$$I_1 = g_{11}V_1 + g_{12}I_2$$

$$V_2 = g_{21}V_1 + g_{22}I_2$$

To find g_{11} and g_{21} , are open-circuit the output port and connect a voltage source V_1 to the input port

Step 2 of 9



Step 3 of 9

s and 1Ω are in series and in parallel to 1Ω resistor.

$$\Rightarrow (s+1) \parallel 1$$

$$\frac{s+1}{s+1+1} = \frac{s+1}{s+2}$$

$$Z = \frac{s+1}{s+2} + s$$

$$Z = \frac{s+1+s^2+2s}{s+2}$$

$$Z = \frac{s^2+3s+1}{s+2}$$

Voltage across the impedance is

$$V_1 = I_1 Z$$

$$V_1 = \left(\frac{s^2+3s+1}{s+2} \right) I_1$$

$$\frac{I_1}{V_1} = \frac{s+2}{s^2+3s+1}$$

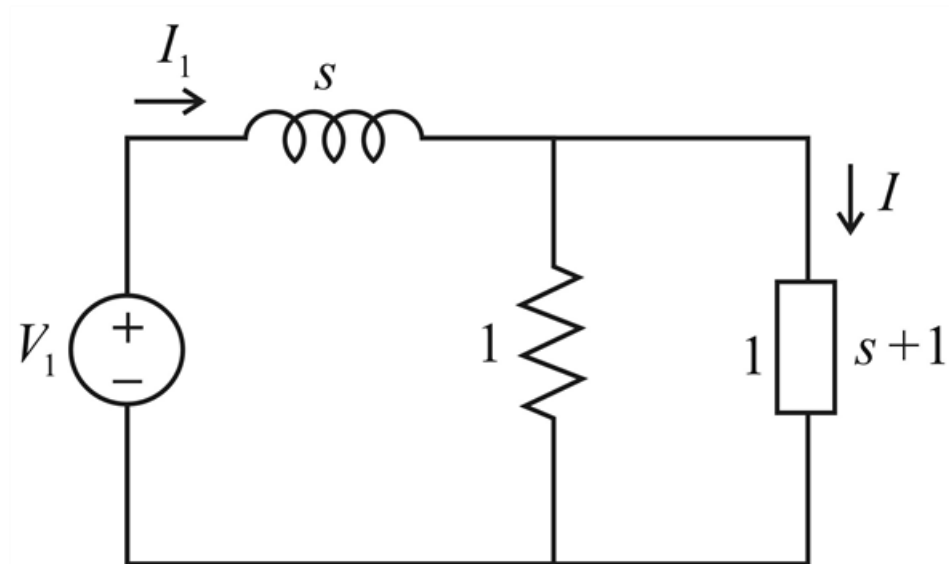
We know that

$$g_{11} = \frac{I_1}{V_1}$$

$$\therefore g_{11} = \frac{s+2}{s^2+3s+1}$$

Thus the circuit can be redrawn as

Step 4 of 9



Step 5 of 9

Using current division we get

$$I = \frac{(1)}{s+1+1} I_1$$

$$I = \frac{1}{s+2} I_1$$

Voltage V_2 across 1Ω resistor is

$$V_2 = 1(I)$$

$$V_2 = 1\left(\frac{1}{s+2}\right) I_1$$

$$V_2 = \left(\frac{1}{s+2}\right) \left(\frac{s+2}{s^2+3s+1}\right) V_1$$

$$\frac{V_2}{V_1} = \frac{1}{s^2+3s+1}$$

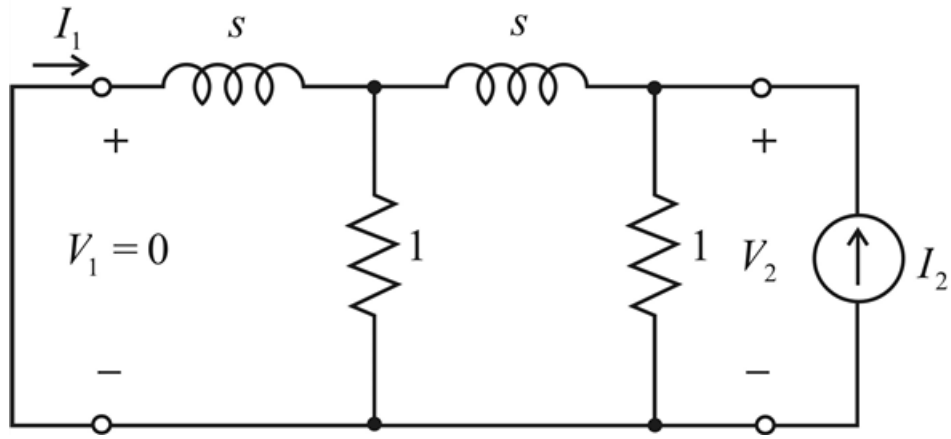
We know that

$$g_{21} = \frac{V_2}{V_1}$$

$$\therefore \boxed{g_{21} = \frac{1}{s^2+3s+1}}$$

Step 6 of 9

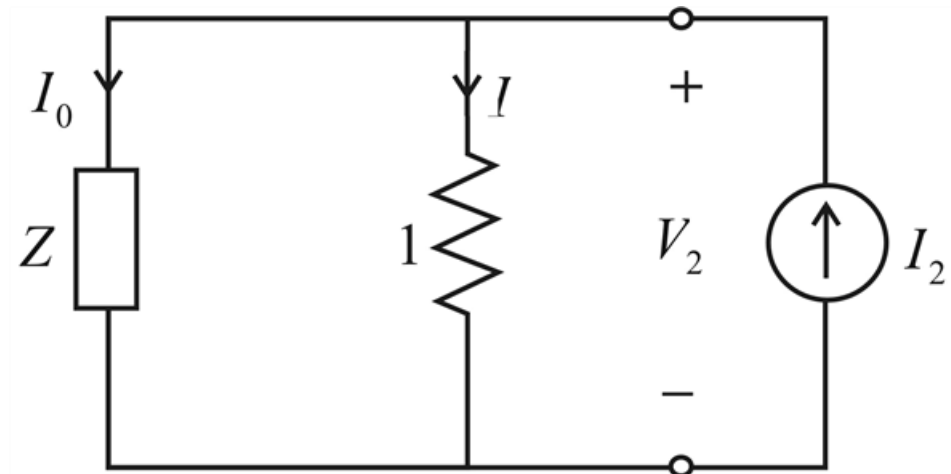
To obtain g_{12} and g_{22} we short circuit the input port and connect a current source I_2 to the output port.



s , 1Ω are in parallel and in series with s then

$$\frac{s \times 1}{s+1} = \frac{s}{s+1}$$

$$Z = \frac{s}{s+1} + s = \frac{s^2 + 2s}{s+1}$$



Using current division we get

$$I = \frac{\frac{s^2 + 2s}{s+1}}{\frac{s^2 + 2s}{s+1} + 1} I_2$$

$$I = \frac{s^2 + 2s}{s^2 + 2s + s + 1} I_2$$

$$I = \frac{s(s+2)}{s^2 + 3s + 1} I_2$$

Where

$$V_2 = (1)I$$

$$V_2 = (1) \left(\frac{s(s+2)}{s^2+3s+1} \right) I_2$$

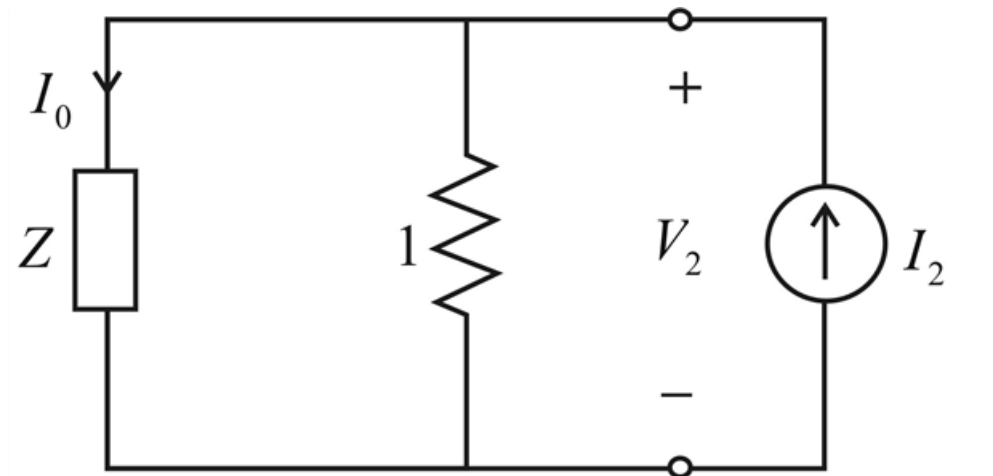
$$\frac{V_2}{I_2} = \frac{s(s+2)}{s^2+3s+1}$$

We know that

$$g_{22} = \frac{V_2}{I_2}$$

$$\therefore g_{22} = \frac{s(s+2)}{s^2+3s+1}$$

Step 8 of 9



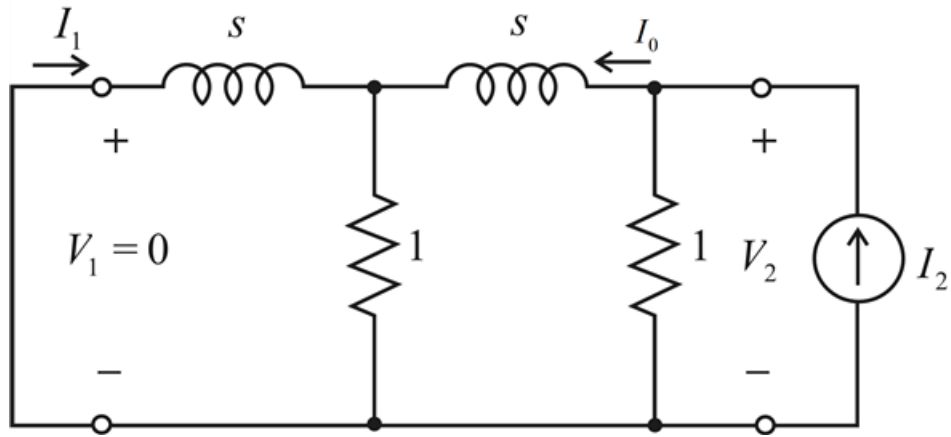
Step 9 of 9

Using current division

$$I_0 = \frac{1}{1+Z} I_2$$

$$I_0 = \frac{1}{1 + \frac{s^2 + 2s}{s+1}} I_2$$

$$I_0 = \frac{s+1}{s^2 + 3s + 1} I_2$$



Hence $I_1 = \frac{-1}{s+1} I_0$

$$I_1 = \left[\frac{-1}{s+1} \right] \left[\frac{s+1}{s^2 + 3s + 1} \right] I_2$$

$$\frac{I_1}{I_2} = \frac{-1}{s^2 + 3s + 1}$$

We know that

$$g_{12} = \frac{I_1}{I_2}$$

$$\therefore g_{12} = \frac{-1}{s^2 + 3s + 1}$$

Thus,

$$[g] = \begin{bmatrix} \frac{s+2}{s^2 + 3s + 1} & \frac{-1}{s^2 + 3s + 1} \\ \frac{1}{s^2 + 3s + 1} & \frac{s(s+2)}{s^2 + 3s + 1} \end{bmatrix}$$